



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 9 • STANDARD 6.AT.11

Write Equations to Represent Relationships Between Two Variables

Lesson 9-3 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

Language Objective: I can define the variables in my equation and explain what each part represents using the words equation, variable, and rate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

equation

variable

rate

represent

define the variables



Tap any word to see what it means and a picture.

1 A mathematical sentence with an equals sign is an .

2 A letter standing for a quantity that can change is a .

3 A fixed amount of one quantity for every one unit of the other is a .

4 To show a relationship with a table, graph, or equation is to it.

5 To state what each letter stands for is to
 .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Would you recommend that Diamond purchase or stream songs from the online music service?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **equation** — A mathematical sentence with an equals sign that shows two amounts are the same.
ecuación — Una oración matemática con un signo igual que muestra que dos cantidades son iguales.

☐ **variable** — A letter that represents a quantity that can change, or vary.
variable — Una letra que representa una cantidad que puede cambiar o variar.

☐ **rate** — A fixed amount of one quantity for every one unit of the other quantity.
tasa — Una cantidad fija de una cantidad por cada unidad de la otra cantidad.

☐ **represent** — To show a relationship using a tool such as a table, a graph, or an equation.
representar — Mostrar una relación usando una herramienta como una tabla, una gráfica o una ecuación.

☐ **define the variables** — To state exactly what quantity each letter in an equation stands for.
definir las variables — Decir exactamente qué cantidad representa cada letra de una ecuación.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

**Diamond should write an ____ for each option so she can ____ the costs.
Each equation uses the ____ as the input and gives the ____ as the output.**

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The fee is charged EVERY month, so the total must be the rate multiplied by the number of months, not added to it once.

Evidence: A strong answer tests the equation against a table row (2 months should give \$49.90) rather than just stating a rule. A weak answer picks multiplication without checking it against the actual numbers.

Reasoning: This evidence connects the definition of equation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Diamond should write an ____ for each option so she can ____ the costs. Each equation uses the ____ as the input and gives the ____ as the output.**
- **My answer is ____.**

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** equation
2. **Notes 2:** variable
3. **Notes 3:** rate
4. **Notes 4:** represent
5. **Notes 5:** define the variables
6. **Watch for this mistake:** *Writing the equation with addition — $c = 24.95 + m$ — instead of multiplication. The fee is charged EVERY month, so the total is the rate times the number of months; always test the equation against a row of the table.*
7. **Try It 1:** A. The equation gives the total cost for ANY number of months, not just the rows listed — *A table can only show the values someone wrote down; substituting any value of m into $c = 24.95m$ gives the cost for that many months instantly.*
8. **Try It 2:** A. 8 hours — *Divide both sides by 15: $h = 120 \div 15 = 8$.*